

AEROSPACE & SPACE TECHNOLOGY

# Anjanisutah 3Space PVT. LTD.

India's Next-Generation Space Technology — Building indigenous, reusable launch systems for the future of Indian aerospace.





# Who We Are

3Space is a pioneering Indian aerospace and space-technology company, dedicated to developing indigenous propulsion systems and fully reusable launch vehicles.

Established in 2025 and incubated at PDEU IIC, our mission begins with an engine-first approach, laying the foundation for a sustainable and cost-effective future in space access.

Founded  
2025

Industry  
Aerospace & Space  
Technology

Focus  
Reusable Launch  
Vehicle Ecosystem

Our vision is to build a scalable Indian reusable launch-vehicle ecosystem, ensuring cost-effective and high-frequency access to space.



OUR STRATEGY & VISION

# Engine-First Approach

3Space is pursuing an **engine-first development approach**, building propulsion technology as the foundation for a future reusable launch-vehicle ecosystem.

Our current core technology is **FFSC MethaLOX propulsion**, designed for efficiency and reusability.

## Our Vision

To build a **scalable Indian reusable launch-vehicle ecosystem** capable of providing cost-effective and high-frequency access to space.

# The Challenge We Are Solving

## Expensive

Small payloads often face disproportionately high launch costs, creating significant barriers to space access for many.

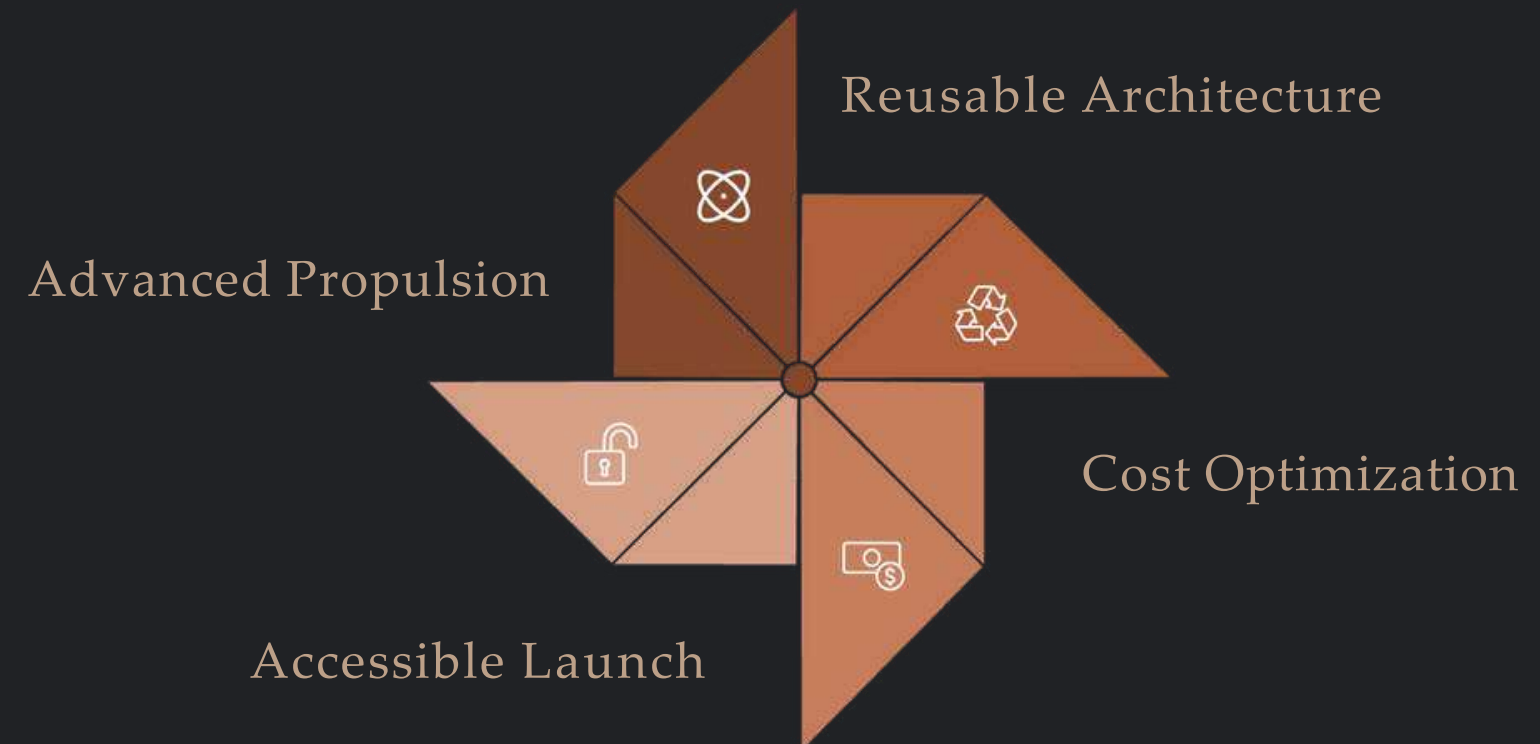
## Infrequent

Limited launch windows and opportunities can severely delay critical research, defence, and commercial missions.

## Limited Reusability

Developing and implementing high-performance, reusable propulsion systems is inherently complex and capital intensive.

## Our Solution



Our strategy is to first develop indigenous propulsion technology, validate it thoroughly, and then progressively integrate it into a reusable launch vehicle. This integrated approach will culminate in providing highly accessible, cost-effective, and frequent launch services.



3SPACE

LAUNCH PROGRAM

# ANJANI-MK1

OUR FIRST LAUNCH-ORIENTED PROJECT

## Objective

To demonstrate and validate technologies forming the foundation of 3Space's future reusable launch-vehicle programme.

## Current Focus

- Vehicle development
- Propulsion integration
- Flight-system development
- Recovery/reusability technologies
- System-level validation

## Target Launch Window

DECEMBER 2026 — FEBRUARY 2027

Development → Integration → Testing → Launch



# KARN-V1

DEVELOPING THE PROPULSION CORE FOR OUR FUTURE REUSABLE ROCKET  
KARN-V1 represents 3Space’s propulsion-development program, focused on creating the advanced engine technology that will form the core of our future reusable orbital launch vehicle.

## Development Focus



Design

Propulsion architecture & subsystem development

Build

Engine hardware & component development

Test

Cold-flow, hot-fire & performance testing

Validate

Reliability, endurance & performance characterization

## Long-Term Role

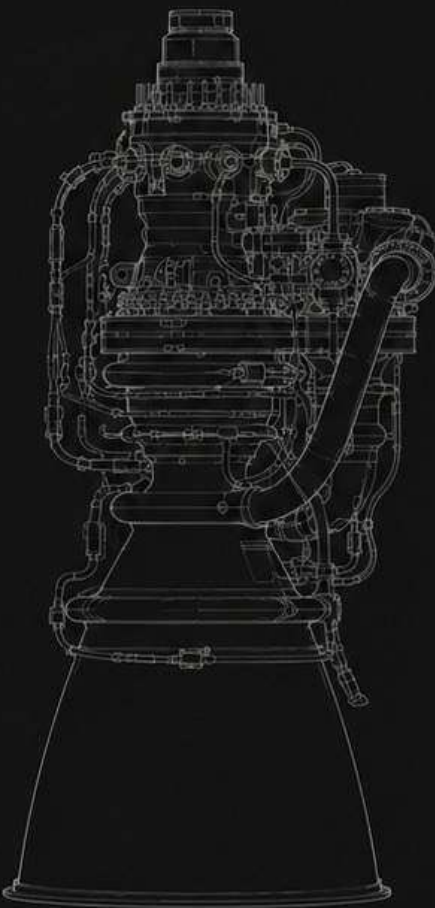
KARN-V1 → Validated Propulsion Core → Reusable Orbital Launch Vehicle



# KV-1

150 kN-class Engine

|                       |                                                                                     |                        |
|-----------------------|-------------------------------------------------------------------------------------|------------------------|
| CYCLE                 |  | FFSC                   |
| PROPELLANTS           |  | LOX / LCH <sub>4</sub> |
| SEA LEVEL THRUST      |  | 150 kN                 |
| REUSABLE              |  | YES                    |
| ENGINE WEIGHT         |  | ~250 kg                |
| THRUST VECTOR CONTROL |  | GIMBAL                 |
| STARTS                |  | MULTIPLE               |
| APPLICATION           |  | BOOSTER                |
| COOLING               |  | REGENERATIVE           |



# Building With the Space & Innovation Ecosystem



## INCUBATED AT

### **PDEU IIC**

Our incubation and entrepreneurship ecosystem supporting 3Space's growth and development.



## SUPPORTED BY

### **iHUB • GOVERNMENT OF GUJARAT • IN-SPACE**

Institutional support helping us advance our technology, innovation and participation in India's growing space ecosystem.



## GRANTS & FUNDING SUPPORT

### **SSIP • iHUB**

Grant support enabling technology development, prototyping and innovation activities.



## ACTIVE PARTICIPATION

### **IN-SPACE ROCKETRY COMPETITION**

Active participation in the IN-SPACE rocketry ecosystem, gaining practical experience in rocketry, engineering and collaborative technology development.

TEAM OVERVIEW

# The People Behind 3Space



Yashraj Patel  
Founder, CEO & CTO



Khushi Piparva  
Chief Propulsion Engineer



Priyank Patel  
Chief Design & Structures Engineer



Aishwarya Raj  
Public Relations Lead



Megha Krishna  
HR Lead

📄 **15+ Member Team** — Comprising 10+ Technical experts and 5+ Management professionals, our team brings a wealth of experience and innovation.

# BUILD THE FUTURE OF SPACE WITH US

At 3Space, we believe the future of space technology is built through collaboration, innovation, and people who are willing to build what has never been built before.

HAVE AN IDEA, OPPORTUNITY OR WANT TO  
BE PART OF THE JOURNEY?

Let's connect.

✉@ General Enquiries:

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✉@ Careers & Opportunities:

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